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ITA0502
Computer Vision
Slot -C

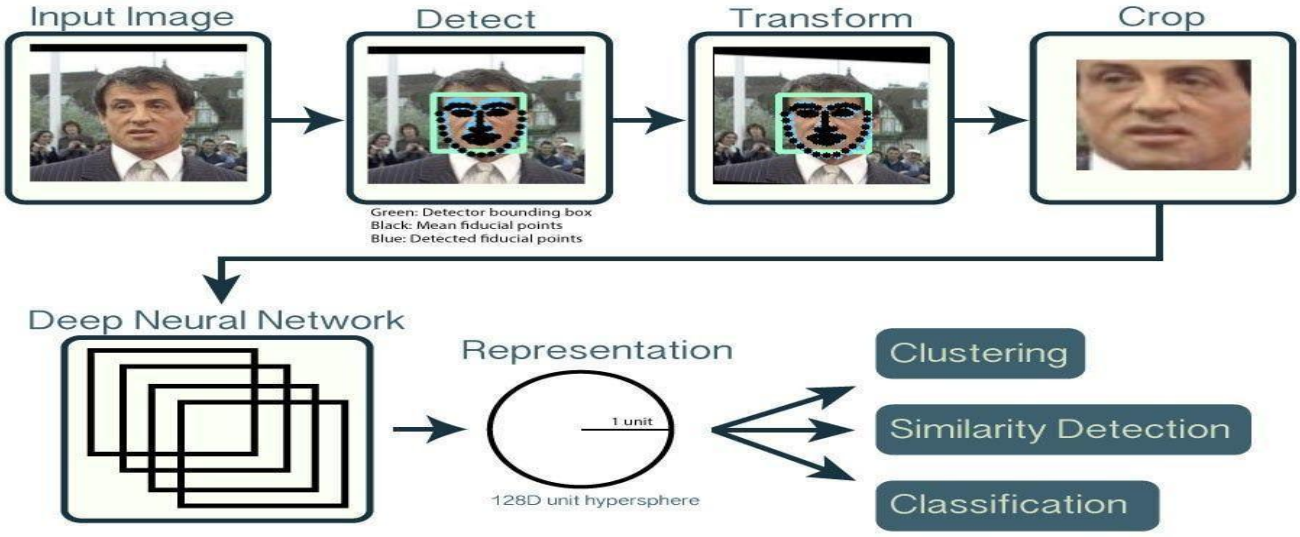
PSEUDOCODE WRITING TASK

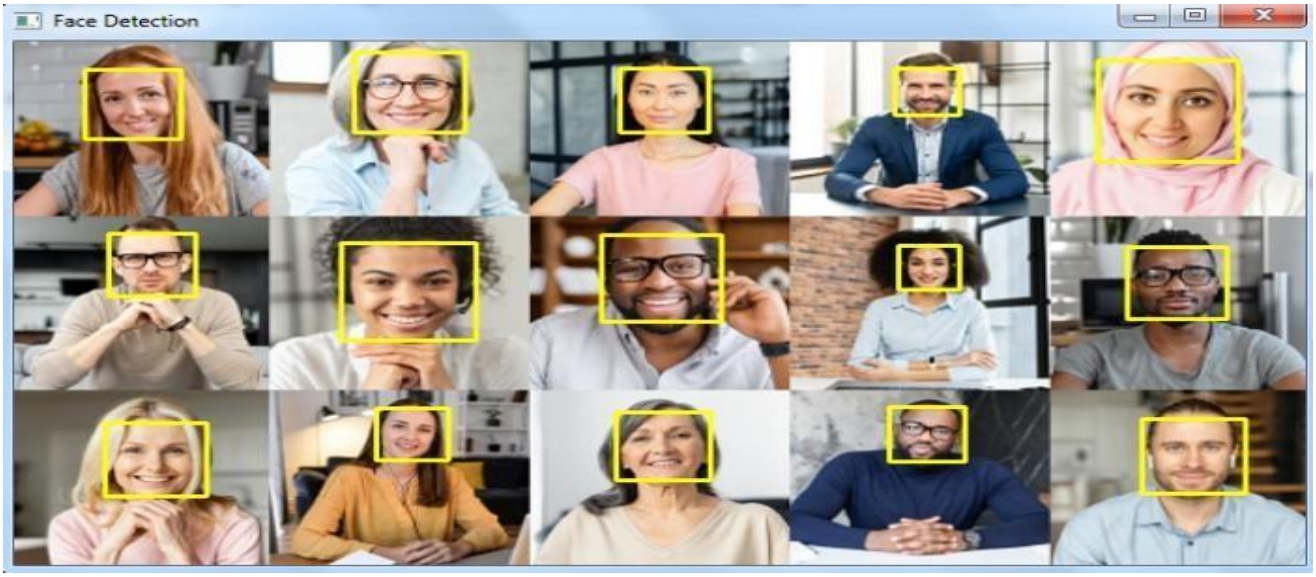
1. Real-Time Face Detection and Recognition

Objective: Design an OpenCV-based system that continuously captures frames from a camera, detects multiple faces, recognizes registered persons, and displays their names with bounding boxes while maintaining real-time performance.

Overall Pipeline:

Camera → Frame Acquisition → Preprocessing → Face Detection → Face Feature Extraction → Recognition → Confidence Check → Visualization → Display





Explanation of Important Steps

BEGIN
1. INITIALIZE SYSTEM
Load OpenCV library
Load trained face detection model
Load trained face recognition dataset/model
Set recognition threshold
Set frame-skip value if required

2. OPEN CAMERA
camera ← Open webcam/CCTV stream
IF camera cannot be opened THEN
Display "Camera Error"
EXIT
END IF

3. START REAL-TIME LOOP
WHILE camera is open DO
3.1 CAPTURE FRAME
frame ← Read frame from camera
IF frame is empty THEN
BREAK
END IF
3.2 OPTIMIZE FRAME
Resize frame to suitable resolution
Convert frame to grayscale
Apply noise reduction if required
3.3 DETECT FACES
faces ← Detect all faces in frame
FOR each detected face DO

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      x, y, width, height ← face coordinates
      3.4 EXTRACT FACE
      face_region ← Crop face from frame
      Resize face_region to required size
      3.5 FEATURE EXTRACTION
      features ← Extract facial features
      OR
      features ← Generate face embedding
      3.6 RECOGNITION
      Compare features with trained dataset
      best_match ← Find closest registered person
      confidence ← Calculate similarity/confidence
      3.7 IDENTITY DECISION
      IF confidence satisfies recognition threshold THEN
        identity ← Registered person's name
      ELSE
        identity ← "Unknown"
      END IF
      3.8 DRAW RESULT
      Draw bounding box around face
      Display identity label above bounding box
    END FOR
    3.9 DISPLAY FRAME
    Show processed frame
    3.10 CHECK EXIT CONDITION
    key ← Read keyboard input
    IF key is ESC OR 'q' THEN
      BREAK
    END IF
  END WHILE

4. RELEASE RESOURCES
  Release camera
  Close all OpenCV windows
  Free memory/resources
END
```

1. Image Acquisition

The camera continuously captures frames.

Camera → Frame 1 → Frame 2 → Frame 3 → ...

OpenCV's VideoCapture can be used to access the webcam or CCTV stream.

2. Preprocessing

Each frame can be:

- Resized to reduce computation.
- Converted to grayscale for lightweight face detection.
- Filtered to reduce camera noise.

This reduces processing time and improves detection quality.

3. Multiple Face Detection

The detector searches the complete frame and returns multiple bounding boxes.

For example:

Face 1 → Student A

Face 2 → Student B

Face 3 → Unknown

Face 4 → Student C

The system processes each detected face independently.

4. Feature Extraction

The detected face is converted into useful numerical features.

For a lightweight OpenCV implementation, **LBPH** can be used.

For a more advanced system, facial embeddings can be generated using a deep facerecognition model.

5. Recognition

The extracted features are compared with the trained dataset.

Detected Face
↓
Feature Extraction
↓
Compare with Dataset
↓
Best Match
↓

Confidence Check

↓

Name / Unknown

6. Confidence Threshold

A threshold is important to reduce false recognition.

IF confidence is sufficient

→ Recognized Person

ELSE

→ Unknown

The exact threshold depends on the selected recognition method.

7. Visualization

The final frame displays:

- Face bounding box.
- Person's identity.
- Unknown label when no reliable match exists.

8. Real-Time Optimization To

minimize processing delay:

- Resize input frames.
- Process a smaller frame resolution.
- Skip some frames if necessary.
- Detect faces periodically.

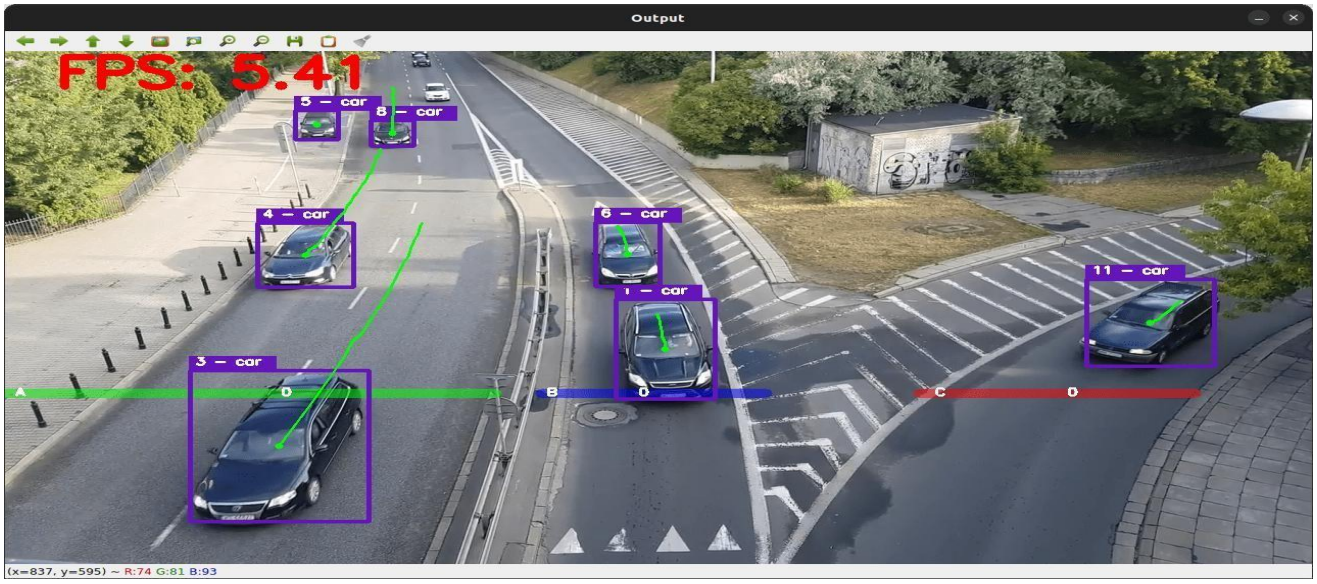
2. Vehicle Detection, Tracking and Counting

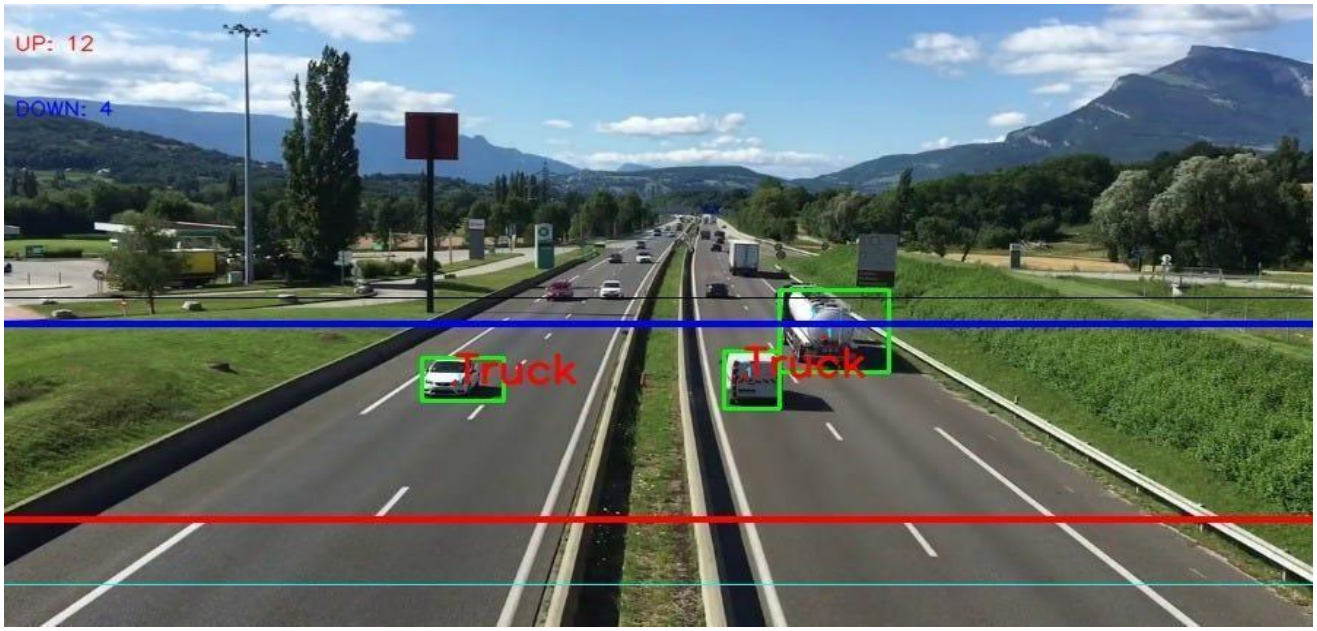
Objective

Design an OpenCV-based traffic monitoring system that detects multiple vehicles, tracks them across frames, counts each vehicle when it crosses a predefined line, and prevents double counting.

Overall Pipeline

Video → Preprocessing → ROI → Vehicle Detection → Tracking → ID Assignment → Line Crossing → Counting → Visualization





```
BEGIN
1. INITIALIZE SYSTEM
   Load OpenCV library
   Load vehicle detection method/model
   Initialize object tracker
   Set Region of Interest (ROI)
   Set virtual counting line
   vehicle_count ← 0
   counted_IDs ← empty set
   next_ID ← 1

2. OPEN VIDEO SOURCE
   video ← Open live camera/video stream
   IF video cannot be opened THEN
      Display "Video Error"
      EXIT
   END IF

3. START PROCESSING LOOP
   WHILE video is open DO
      3.1 READ FRAME
      frame ← Read frame
      IF frame is empty THEN
         BREAK
      END IF

      3.2 PREPROCESS FRAME
      Resize frame if required
      Apply Gaussian/median filtering
      Define/use ROI
      roi_frame ← Extract ROI from frame

      3.3 DETECT VEHICLES
      detections ← Detect vehicles in ROI
      FOR each detection DO
         Obtain bounding box
         Obtain vehicle class
         Obtain confidence
         IF confidence is below threshold THEN
```

```
      Ignore detection
    END IF
  END FOR
  3.4 UPDATE TRACKERS
  Send valid detections to tracker
  tracker updates vehicle positions
  FOR each tracked vehicle DO
    Obtain tracking ID
    Obtain current bounding box
    Calculate center point
  END FOR
  3.5 CHECK VIRTUAL LINE CROSSING
  FOR each tracked vehicle DO
    current_position ← Current center point
    previous_position ← Previous center point
    IF vehicle moved from one side of virtual line to the other THEN
      IF tracking ID is NOT in counted_IDs THEN
        vehicle_count ← vehicle_count + 1
        Add tracking ID to counted_IDs
      END IF
    END IF
    Store current position as previous position
  END FOR
  3.6 VISUALIZATION
  Draw ROI boundary
  Draw virtual counting line
  FOR each tracked vehicle DO
    Draw bounding box
    Display tracking ID
    Display vehicle class if required
    Draw center point
  END FOR
  Display vehicle_count
  3.7 DISPLAY FRAME
  Show processed frame
  3.8 CHECK EXIT CONDITION
  key ← Read keyboard input
  IF key is ESC OR 'q' THEN
    BREAK
  END IF
END WHILE
```

4. RELEASE RESOURCES
   Release video source
   Destroy all OpenCV windows
   Release tracker resources
END

1. Video Acquisition

The system continuously reads frames from:

- CCTV camera
- Webcam
- Traffic surveillance camera
- Recorded traffic video

Camera
↓
Frame 1
↓
Frame 2
↓
Frame 3
↓

If the video source cannot be opened, the program terminates safely.

2. Preprocessing The

input frame can be:

- Resized.
- Smoothed using Gaussian/median filtering.
- Restricted to a **Region of Interest**.

The ROI prevents the system from processing irrelevant areas such as buildings, sky, and sidewalks.

3. Vehicle Detection

The detector identifies vehicles inside the ROI.

Possible approaches include:

- Background subtraction + contours for a simple fixed-camera system.
- OpenCV DNN/object detection for more reliable vehicle detection.

The detector returns:

Bounding Box

Vehicle Class Confidence

Score

Example:

Car → Confidence 95%

Bus → Confidence 91%

Truck → Confidence 88%

Low-confidence detections are ignored.

4. Vehicle Tracking

Detection alone is not sufficient because the same vehicle appears in many consecutive frames.

Tracking assigns an ID:

Frame 1 → Car → ID 1

Frame 2 → Car → ID 1

Frame 3 → Car → ID 1

Frame 4 → Car → ID 1

Therefore, the system understands that these detections belong to the **same vehicle**.

6. Line-Crossing Logic

Suppose the counting line is at position Y.

If:

Previous Y < Line Y

AND

Current Y >= Line Y then the

vehicle has crossed the line.

The counter is increased:

vehicle_count = vehicle_count + 1

7. Avoiding Double Counting

This is one of the most important constraints.

Maintain a set:

counted_IDs = {1, 4, 7}

Before increasing the count: IF

vehicle_ID NOT IN counted_IDs

count = count + 1

Add vehicle_ID to counted_IDs

END IF

Therefore:

Vehicle ID 1 crosses line → Count = 1

Vehicle ID 1 appears again → Do NOT count

Vehicle ID 1 appears again → Do NOT count

This prevents duplicate counting.

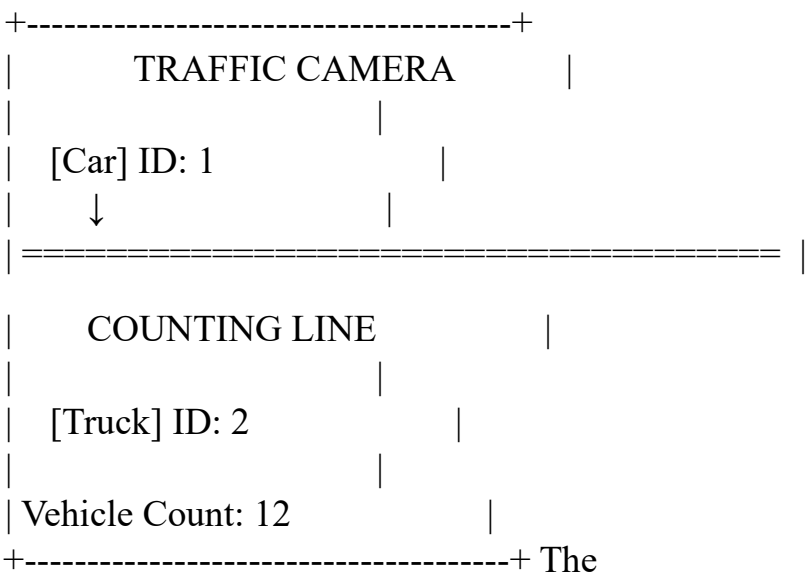
8. Multiple Vehicles

Each vehicle has its own:

- Bounding box
- Tracking ID
- Center position
- Previous position
- Counting status

9. Visualization

The output frame displays:



The following information can be displayed:

- Bounding boxes.
- Vehicle IDs.
- Vehicle classes.
- Virtual line.
- ROI.
- Total vehicle count.

10. Real-Time Optimization To

maintain near real-time performance:

- Resize frames.
- Process only the ROI.
- Use efficient detection models.
- Track vehicles between detection frames.
- Skip unnecessary frames when appropriate.
- Remove expired tracking IDs.
- Avoid repeatedly detecting the entire image.